

FlowMate

Product Requirements Document (PRD) v2

Schedule Debugger Edition

Core Positioning

FlowMate is not another calendar. It is a recovery engine that helps students debug a broken day and save what matters most.

Product Name	FlowMate
Product Concept	Schedule Debugger for chaotic student days
Primary Category	Productivity + Wellness
Target Platform	Responsive web app deployed on Google Cloud Run
AI Engine	Gemini via Google AI Studio / Gemini API
Document Version	v2.0

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Product Identity

Product name: FlowMate

Product concept: Schedule Debugger for chaotic student days

Tagline: Debug your broken day. Recover what matters.

Indonesian tagline: Bukan membuat hari sempurna, tetapi menyelamatkan hal yang paling penting.

Primary category: Productivity + Wellness

Competition alignment: #JuaraVibeCoding / Gemini + Google AI Studio + Cloud Run

Document version: v2.0 - Schedule Debugger Edition

1. Executive Summary

FlowMate adalah aplikasi web berbasis AI yang membantu mahasiswa menyelamatkan jadwal harian yang sudah berantakan. FlowMate tidak diposisikan sebagai kalender biasa atau sekadar auto-scheduler. FlowMate diposisikan sebagai Schedule Debugger: sistem yang mendeteksi konflik, membaca energi pengguna, memisahkan event wajib dan fleksibel, lalu menghasilkan recovery plan yang realistis.

Masalah utama yang diselesaikan FlowMate adalah momen ketika pengguna merasa harinya sudah gagal: bangun kesiangan, energi rendah, tugas menumpuk, kelas wajib tetap harus hadir, dan jadwal lama tidak lagi masuk akal. Dalam momen itu, aplikasi kalender biasa hanya menunjukkan jadwal yang rusak. FlowMate bertugas memperbaiki jadwal tersebut.

Nilai utama FlowMate adalah human recovery scheduling: membuat ulang rencana hari ini berdasarkan kondisi nyata pengguna, bukan berdasarkan asumsi bahwa pengguna selalu disiplin, selalu sehat, dan selalu punya energi penuh.

2. Product Vision

FlowMate ingin menjadi AI recovery assistant untuk mahasiswa dan pekerja kreatif yang hidupnya tidak selalu rapi. Produk ini membantu pengguna berpindah dari kondisi panic, guilt, dan overwhelm menuju kondisi yang lebih terarah: apa yang harus tetap dilakukan, apa yang bisa dipindahkan, apa yang perlu dibatalkan, dan kapan tubuh perlu istirahat.

Visi produk:

FlowMate turns a broken day into a realistic recovery plan.

Dalam konteks engineering, FlowMate melihat jadwal sebagai constraint optimization problem. Setiap event memiliki durasi, prioritas, deadline, status wajib atau fleksibel, kebutuhan energi, dan batas waktu. Ketika pengguna bangun terlambat atau energi menurun, jadwal lama menjadi infeasible. FlowMate mencari konfigurasi baru yang feasible, manusiawi, dan mudah dieksekusi.

3. Competition Context - #JuaraVibeCoding

FlowMate dirancang untuk cocok dengan semangat #JuaraVibeCoding: menggunakan bahasa natural, Gemini, Google AI Studio, dan Cloud Run untuk membangun aplikasi yang nyata, deployed, dan memiliki value personal.

Checklist kompetisi yang ditargetkan:

- Aplikasi aktif dan dapat diakses melalui Google Cloud Run.
- Gemini digunakan sebagai AI scheduling and recovery engine.
- Demo video 2-3 menit dipublikasikan di LinkedIn secara publik.
- Unggahan menggunakan hashtag #JuaraVibeCoding.
- Link final dikirim melalui formulir penyelesaian resmi.

Kategori yang paling cocok:

- Produktivitas: FlowMate mengelola jadwal yang berantakan dan menghemat waktu rescheduling manual.

- Kesejahteraan / Wellness: FlowMate mempertimbangkan energi, tidur, dan beban mental pengguna.
- Kehidupan sehari-hari: FlowMate menyelesaikan masalah harian yang sering dialami mahasiswa.

4. Target Users

4.1 Primary Target User

Mahasiswa Indonesia dengan beban akademik tinggi, khususnya bidang STEM, IT, robotika, dan teknik, yang sering mengalami jadwal kacau karena begadang, praktikum, kelas wajib, tugas kelompok, organisasi, dan deadline mendadak.

4.2 User Persona

Nama persona: Raka, mahasiswa teknik robotika.

Kondisi harian:

- Sering tidur lewat tengah malam karena tugas coding atau robotik.
- Jadwal kelas dan praktikum tidak selalu fleksibel.
- Punya deadline LMS, tugas kelompok, organisasi, dan latihan proyek.
- Sering merasa gagal ketika satu jadwal terlewat.
- Butuh sistem yang tidak menghakimi, tetapi membantu menyelamatkan sisa hari.

4.3 Jobs-to-be-Done

Ketika jadwal saya rusak, saya ingin aplikasi membantu memilih ulang prioritas hari ini, sehingga saya tetap bisa menyelesaikan hal terpenting tanpa panik.

Ketika energi saya rendah, saya ingin jadwal saya menyesuaikan kapasitas tubuh saya, bukan memaksa saya mengikuti rencana yang tidak realistis.

Ketika saya melihat jadwal baru, saya ingin tahu alasan AI mengubahnya, sehingga saya bisa percaya pada keputusan sistem.

5. Problem Statement

Aplikasi kalender tradisional berasumsi bahwa pengguna akan mengikuti jadwal sesuai rencana. Masalahnya, hidup mahasiswa tidak selalu linear. Satu event yang terlewat dapat membuat jadwal berikutnya bertumpuk. Dalam kondisi energi rendah, pengguna tidak hanya kehilangan waktu, tetapi juga kehilangan kemampuan mengambil keputusan.

Masalah utama:

1. Schedule fatigue: satu blok waktu terlewat membuat jadwal seharian terasa gagal.
2. Cognitive overload: memindahkan banyak event secara manual membutuhkan energi mental besar.
3. Lack of human context: kalender tidak tahu pengguna tidur 3 jam, belum makan, atau sedang stres.
4. Guilt loop: pengguna merasa gagal, lalu semakin sulit memulai ulang hari.
5. Generic productivity tools: banyak aplikasi fokus pada efisiensi, bukan recovery.

Engineering framing:

Sebuah jadwal disebut feasible jika semua event penting dapat ditempatkan tanpa overlap, sesuai constraint waktu, prioritas, dan energi. Ketika input dunia nyata berubah, jadwal lama perlu di-debug.

6. Differentiation Strategy

FlowMate tidak bersaing langsung sebagai calendar automation untuk pekerja korporat. FlowMate mengambil posisi berbeda:

Not another calendar. A recovery engine for chaotic days.

Aplikasi sejenis biasanya mengoptimalkan jadwal berdasarkan task, deadline, dan available time. FlowMate menambahkan lapisan baru: kondisi manusia. Sistem tidak hanya bertanya "kapan deadline?", tetapi juga "apakah kamu punya energi untuk mengerjakannya hari ini?"

6.1 Differentiation Matrix

Product Type	Primary Focus	Gap	FlowMate Angle
Traditional calendar	Menyimpan event	Tidak membantu ketika jadwal rusak	Rescue plan otomatis
Auto-scheduler	Menaruh task ke slot kosong	Terlalu task-centric	Energy-aware recovery
To-do app	Mengelola daftar tugas	Tidak paham konflik waktu	Calendar diff and repair
Wellness app	Refleksi dan mood tracking	Tidak langsung memperbaiki jadwal	Mood and energy influence schedule
FlowMate	Recovery scheduling	Fokus pada hari yang sudah kacau	Schedule Debugger

7. Core Value Proposition

FlowMate membantu pengguna menyelamatkan hari yang berantakan dengan tiga hasil nyata:

1. Conflict reduction: konflik jadwal berkurang.
2. Priority preservation: event wajib tetap aman.
3. Energy fit: tugas berat ditempatkan pada slot yang lebih realistis.

Value output yang harus terlihat di UI:

- Berapa event penting yang berhasil diselamatkan.
- Berapa konflik yang dihapus.
- Berapa tugas yang dipindahkan.
- Berapa blok recovery yang ditambahkan.
- Recovery Score sebelum dan sesudah reschedule.

Contoh:

Before: Recovery Score 31%. After: Recovery Score 84%.

8. Product System Concept - Schedule Debugger

FlowMate memperlakukan jadwal seperti program yang error. Jika jadwal memiliki overlap, event terlewat, energi tidak cukup, atau deadline terlalu dekat, maka sistem melakukan debugging.

Mental model UI:

- Detecting conflicts...
- Analyzing energy budget...
- Protecting fixed events...
- Generating recovery plan...
- Schedule patched successfully.

Sistem utama:

1. Day Damage Detector
2. Energy Budget System
3. Recovery Scheduler
4. AI Decision Log
5. Visual Diff Calendar
6. Pattern Insight Engine

9. Core Features

9.1 One-Tap Rescue / Debug My Day

Deskripsi:

Fitur utama berupa tombol besar untuk menyelamatkan jadwal yang sudah berantakan.

User flow:

1. User menekan tombol "Debug My Day" atau "Jadwalku Berantakan".
2. User memasukkan kondisi dalam bahasa natural, teks atau voice.
3. Sistem membaca jadwal hari ini dan event 7 hari ke depan.
4. Sistem mendeteksi event yang terlewat, konflik, event wajib, dan tugas fleksibel.
5. Gemini menghasilkan recovery plan dalam format JSON.
6. Backend memvalidasi output AI.
7. UI menampilkan Before vs After Calendar.
8. User melihat AI Decision Log.
9. User memilih Apply atau Reject.

Contoh input:

Aku baru bangun jam 12. Energi 40%. Kelas jam 2 wajib. Tugas robotika deadline malam ini. Gym boleh dibatalkan.

Contoh output:

- Kuliah jam 14.00 dipertahankan.
- Makan dan recovery block ditambahkan.
- Gym dibatalkan.
- Tugas robotika dipindahkan ke slot malam.
- Tugas ringan digeser ke besok.

9.2 Day Damage Detector

Deskripsi:

Modul yang menghitung tingkat kerusakan jadwal hari ini.

Input:

- Current time
- Missed events
- Remaining available time
- Fixed events
- User energy level
- Deadline pressure
- Number of overlapping events

Output:

- Damage Score 0-100
- Damage Level: low, medium, high, critical
- Summary of schedule problems

Formula konseptual:

$$\text{damage_score} = \text{missed_weight} + \text{overlap_weight} + \text{deadline_pressure} + \text{low_energy_penalty} - \text{available_recovery_time}$$

Contoh UI:

Damage Level: 73% - Jadwal lama tidak realistis. Recovery plan diperlukan.

9.3 Energy Budget System

Deskripsi:

FlowMate memberi bobot energi untuk setiap aktivitas. Sistem memastikan jadwal baru tidak hanya muat secara waktu, tetapi juga masuk akal secara kapasitas pengguna.

Contoh energy model:

Activity Type	Energy Cost / Gain
Kuliah wajib	-25
Deep work	-35
Tugas ringan	-10
Meeting / organisasi	-20
Gym	-30
Makan	+10
Power nap	+20
Short break	+8

Rule:

Jika energy budget pengguna rendah, sistem tidak boleh menaruh aktivitas berat secara berurutan tanpa recovery block.

9.4 Recovery Scheduler

Deskripsi:

Modul yang membuat ulang jadwal berdasarkan constraint.

Objective function:

```

maximize:
    priority_completed + fixed_events_preserved + energy_fit + deadline_safety

minimize:
    time_overlap + cognitive_load + unrealistic_tasks + unnecessary_context_switching

```

Constraint utama:

- Event fixed tidak boleh dihapus.
- Event flexible boleh digeser.
- Event optional boleh dibatalkan.
- Tugas berat membutuhkan energy threshold tertentu.
- Recovery block wajib ditambahkan jika energi rendah.
- Tidak boleh ada overlap pada final schedule.
- Semua perubahan harus dikonfirmasi user sebelum diterapkan.

9.5 AI Decision Log

Deskripsi:

Panel penjelasan yang menampilkan alasan perubahan jadwal. Panel ini bukan chain-of-thought mentah, tetapi explanation layer yang aman dan mudah dipahami pengguna.

Contoh:

- Kuliah jam 14.00 dipertahankan karena event ini wajib.
- Gym dibatalkan karena energi hanya 40% dan tugas robotika lebih prioritas.
- Tugas robotika dipindahkan ke 19.30 karena membutuhkan fokus tinggi.
- Istirahat 25 menit ditambahkan sebelum kelas untuk menurunkan beban kognitif.

Value:

Meningkatkan trust, memberi transparansi, dan membuat AI terasa seperti partner, bukan black box.

9.6 Before/After Visual Diff Calendar

Deskripsi:

Komponen visual yang menunjukkan perubahan jadwal sebelum dan sesudah rescue.

Before state:

- Event merah: missed, conflict, unrealistic
- Event kuning: risky
- Event abu-abu: optional / can be moved

After state:

- Event hijau: feasible
- Event biru: fixed event preserved
- Event ungu: recovery block
- Event abu-abu transparan: deferred to tomorrow

Wow factor:

Kalender terlihat kacau pada awalnya, lalu berubah menjadi rapi dalam satu klik.

9.7 Smart Morning Check-in

Deskripsi:

Check-in pagi untuk mengurangi kemungkinan jadwal rusak sejak awal.

Input:

- Jam tidur
- Jam bangun
- Mood
- Energy level 1-10
- Top priority hari ini

Output:

- Energy-aware plan
- Suggested focus window
- Warning jika jadwal terlalu berat

9.8 Empathic AI Journal

Deskripsi:

Jurnal harian yang menjadi feedback loop untuk sistem penjadwalan.

Input:

- Keluhan
- Pencapaian
- Hambatan
- Pola energi

AI processing:

- Sentiment analysis
- Entity extraction
- Pattern detection

Output:

- Weekly insight
- Recommendation for future schedule

Contoh insight:

Kamu lebih produktif pada Selasa pagi. FlowMate menyarankan menjadikan slot tersebut sebagai deep work permanen.

9.9 Dashboard Kesejahteraan dan Produktivitas**Deskripsi:**

Dashboard yang menampilkan metrik sederhana dan actionable.

Metrik utama:

- Recovery Score trend
- Energy vs completion rate
- Number of rescued days
- Most common schedule failure reason
- Reschedule rate
- Fixed event preservation rate

10. MVP Scope**10.1 Must Have for Competition MVP**

- Custom web UI with coding-vibe visual identity.
- One-Tap Rescue / Debug My Day.
- Demo calendar dataset.
- User input for energy, mood, and situation.
- Gemini-based recovery plan generation.
- Structured JSON output.
- Backend validation of schedule plan.
- Before/After Calendar Diff.
- AI Decision Log.
- Recovery Score.
- Deploy to Cloud Run.

10.2 Should Have

- Google Calendar read integration.
- Morning Check-in.
- Journal with local persistence.
- Dashboard metrics.
- Demo Mode and Live Mode toggle.

10.3 Could Have

- Voice input.
- Weekly insight.
- Undo / rollback applied changes.
- Mobile PWA install prompt.
- Local notification.

10.4 Out of Scope for MVP

- Full two-way Google Calendar sync.
- Automatic background scheduling without user approval.
- Clinical mental health advice.

- Multi-user collaboration.
- Complex recurring event editing.

11. Functional Requirements

ID	Requirement	Priority
FR-01	User can trigger Debug My Day flow	Must
FR-02	User can input current condition in natural language	Must
FR-03	User can input energy level	Must
FR-04	System can load demo calendar data	Must
FR-05	System can classify events as fixed, flexible, optional, or recovery	Must
FR-06	Gemini returns structured recovery plan JSON	Must
FR-07	Backend validates no-overlap final schedule	Must
FR-08	UI shows Before/After Calendar Diff	Must
FR-09	UI shows AI Decision Log	Must
FR-10	UI shows Recovery Score before and after	Must
FR-11	User can approve or reject proposed schedule	Must
FR-12	System saves rescue session history	Should
FR-13	User can perform Morning Check-in	Should
FR-14	User can write journal entry	Should
FR-15	Dashboard shows basic metrics	Should

12. Non-Functional Requirements

Performance:

- Demo Mode response target: under 5 seconds.
- Live AI response target: under 10 seconds.
- Frontend initial load target: under 3 seconds on average mobile connection.

Reliability:

- Demo Mode must work without Google OAuth.
- If Gemini API fails, show fallback static rescue plan.
- If Calendar API fails, continue using demoCalendar.

Security and Privacy:

- User must approve changes before any calendar update.
- Journal content should not be publicly exposed.
- Production version should avoid storing sensitive tokens in localStorage.
- API keys must be stored in environment variables.

Accessibility:

- UI must support keyboard navigation.
- Color indicators must also include labels, not color only.
- Text contrast must be readable in dark mode.

Ethical Boundaries:

- FlowMate is not a medical or therapy product.

- App language must avoid blaming the user.
- AI suggestions are planning assistance, not clinical advice.

13. Technical Architecture

13.1 Recommended Stack

Frontend:

- Vanilla HTML/CSS/JavaScript or lightweight framework.
- Coding-themed dark UI.
- Responsive layout for mobile and desktop.

Backend:

- FastAPI.
- AI orchestration service.
- Schedule validation service.
- Database service.

AI Engine:

- Gemini via Google AI Studio / Gemini API.
- Structured JSON output.
- Prompt uses constraints, event list, energy, and user context.

Database:

- SQLite + SQLAlchemy for MVP demo.
- Future production migration: Cloud SQL PostgreSQL or Firestore.

Deployment:

- Dockerized web app.
- Google Cloud Run.

13.2 Architecture Diagram

```

Frontend
-> Debug My Day UI
-> Morning Check-in
-> Before/After Calendar
-> AI Decision Log
-> Recovery Score

FastAPI Backend
-> Auth Service
-> Calendar Service
-> Constraint Builder
-> Gemini Planner
-> Schedule Validator
-> Action Logger

Database
-> users
-> checkins
-> journals
-> calendar_snapshots
-> rescue_sessions
-> schedule_actions

External Services
-> Gemini API

```

-> Google Calendar API
-> Cloud Run

14. AI Scheduling Logic

14.1 Input to Gemini

Gemini receives:

- User situation text
- Energy level
- Mood
- Current time
- Calendar events
- Event labels: fixed, flexible, optional
- Journal/check-in summary
- Product rules

14.2 Output from Gemini

Gemini must return structured JSON only.

Required fields:

- summary
- damage_assessment
- energy_assessment
- schedule_actions
- new_schedule
- decision_log
- recovery_score_before
- recovery_score_after
- risk_flags

14.3 Backend Validation

The backend must validate:

- JSON schema is valid.
- Time format is valid.
- No overlap exists in final schedule.
- Fixed events are preserved.
- Event duration is reasonable.
- Recovery block is included when energy is low.
- No destructive calendar action occurs without approval.

15. Data Model

15.1 Tables

users:

- id
- email
- display_name
- created_at

checkins:

- id
- user_id
- sleep_start
- sleep_end
- mood
- energy_level
- top_priority
- created_at

journals:

- id
- user_id
- content
- sentiment_label
- extracted_entities
- created_at

calendar_snapshots:

- id
- user_id
- source
- raw_events_json
- created_at

rescue_sessions:

- id
- user_id
- situation_text
- damage_score_before
- recovery_score_after
- ai_summary
- created_at

schedule_actions:

- id
- rescue_session_id
- action_type
- event_title
- old_start
- old_end
- new_start
- new_end
- reason

16. UX and Visual Direction

FlowMate harus terasa seperti developer tool untuk kehidupan sehari-hari.

Visual identity:

- Dark terminal aesthetic.
- Monospace typography for logs.
- Glowing calendar blocks.
- Animated diff between broken and recovered schedule.
- Command-palette style input.
- System log microcopy.

- Glassmorphism panels with high readability.

Core screen layout:

- Left: calendar timeline.
- Center: rescue input and recovery score.
- Right: AI Decision Log / schedule patch notes.

Hero microcopy:

"Your day crashed. FlowMate can patch it."

Indonesian version:

"Harimu crash? FlowMate bantu patch jadwalmu."

Button labels:

- Debug My Day
- Rescue Schedule
- Apply Patch
- Reject Patch
- View Decision Log

17. Key User Flows

17.1 Rescue Flow

1. User opens app.
2. User sees current schedule and conflict indicators.
3. User clicks Debug My Day.
4. User inputs situation and energy.
5. App runs schedule analysis.
6. App shows damage score.
7. App generates recovery plan.
8. User compares before and after.
9. User reads decision log.
10. User applies or rejects schedule patch.

17.2 Morning Check-in Flow

1. User opens app in the morning.
2. User enters sleep, mood, energy, and top priority.
3. App evaluates whether today's schedule is realistic.
4. App suggests minor adjustments before the day breaks.

17.3 Journal Insight Flow

1. User writes a short journal at night.
2. App extracts patterns and emotional context.
3. Weekly insight engine summarizes patterns.
4. Future schedule suggestions become more personalized.

18. API Endpoint Draft

```
GET  /api/health
GET  /api/demo-calendar
POST /api/rescue
```

```

POST /api/validate-schedule
POST /api/apply-schedule
POST /api/checkin
POST /api/journal
GET /api/dashboard

```

Example request for /api/rescue:

```

{
  "situation": "Aku baru bangun jam 12. Energi 40%. Kelas jam 2 wajib.",
  "energy_level": 4,
  "mood": "overwhelmed",
  "mode": "demo"
}

```

Example response:

```

{
  "summary": "Hari ini disusun ulang agar event wajib tetap aman.",
  "damage_score": 73,
  "recovery_score_before": 31,
  "recovery_score_after": 84,
  "schedule_actions": [
    {
      "action": "move",
      "event_title": "Tugas Robotika",
      "from": "13:00",
      "to": "19:30",
      "reason": "Butuh fokus tinggi dan slot siang sudah tidak realistis."
    }
  ],
  "decision_log": [
    "Kuliah jam 14.00 dipertahankan karena wajib.",
    "Gym dibatalkan karena energi rendah.",
    "Istirahat 25 menit ditambahkan sebelum kelas."
  ]
}

```

19. Demo Strategy

Demo utama harus stabil dan tidak bergantung penuh pada OAuth.

Demo Mode:

- Menggunakan demoCalendar statis.
- Tidak membutuhkan login Google.
- Tetap menampilkan AI plan, visual diff, decision log, dan recovery score.

Live Mode:

- Google OAuth.
- Read Calendar events.
- Generate draft recovery plan.
- Write-back hanya setelah user approval.

Fallback:

- Jika Gemini gagal, tampilkan predefined rescue plan.
- Jika Calendar API gagal, gunakan demoCalendar.
- Jika database gagal, gunakan in-memory session state untuk demo.

20. Demo Video Scenario

Durasi target: 2-3 menit.

Storyboard:**0:00-0:15** - Problem hook:

Mahasiswa bangun jam 12. Jadwal hari ini kacau. Kelas jam 14.00 wajib.

0:15-0:35 - Product reveal:

FlowMate tampil sebagai Schedule Debugger.

0:35-1:00 - Input:**User mengetik:** "Aku baru bangun jam 12, energi 40%, kelas jam 2 wajib, tugas robotika deadline malam ini."**1:00-1:35** - AI processing:

Terminal log menunjukkan conflict detection, energy analysis, and recovery planning.

1:35-2:10 - Wow moment:

Before calendar merah dan overlap berubah menjadi After calendar hijau dan rapi.

2:10-2:35 - Value proof:

Recovery Score naik dari 31% ke 84%. Event wajib aman. Konflik hilang.

2:35-2:55 - Closing:

Tampilkan Cloud Run URL, hashtag #JuaraVibeCoding, dan CTA.

21. Success Metrics

Product metrics:

- Number of rescue sessions created.
- Average recovery score improvement.
- Percentage of fixed events preserved.
- Number of conflicts removed.
- Number of days rescued.
- User acceptance rate of suggested schedule.

UX metrics:

- Time to generate rescue plan.
- Time from panic input to clear plan.
- Number of clicks from app open to result.
- User comprehension of AI Decision Log.

Competition demo metrics:

- Cloud Run app loads successfully.
- Demo completes without manual code changes.
- AI output appears in structured UI.
- Visual diff is clear within 5 seconds.

22. Risks and Mitigations

Risk	Impact	Mitigation
Gemini returns invalid JSON	UI breaks	Use schema validation and fallback response
Google OAuth fails during demo	Demo blocked	Demo Mode with static calendar
Calendar write modifies real events incorrectly	User trust issue	Approval step and write-back disabled by default in MVP
SQLite persistence issue on Cloud Run	Data loss	Use SQLite only for demo; migrate to Cloud SQL/Firestore for production
Product sounds like medical advice	Compliance risk	Position as planning assistant, not

		therapy product
UI too complex	User confusion	Keep rescue flow as the main path
AI suggests unrealistic plan	Poor trust	Backend validation and energy budget constraints

23. Roadmap

Phase 1 - Competition MVP

- Build coding-vibe frontend.
- Implement Debug My Day.
- Integrate Gemini structured output.
- Show visual diff and decision log.
- Deploy to Cloud Run.
- Record LinkedIn video.

Phase 2 - Live Calendar Beta

- Add Google OAuth.
- Read Google Calendar events.
- Apply schedule changes after approval.
- Add undo capability.

Phase 3 - Personalization

- Journal sentiment analysis.
- Weekly pattern insight.
- Personalized energy prediction.

Phase 4 - Production Hardening

- Migrate database to managed storage.
- Improve security and token handling.
- Add PWA support.
- Add monitoring and analytics.

24. Acceptance Criteria

The MVP is accepted if:

- User can open the deployed Cloud Run URL.
- User can trigger Debug My Day.
- User can enter a natural language situation.
- App generates a new schedule using Gemini or fallback.
- App shows Before/After Calendar Diff.
- App shows AI Decision Log.
- App shows Recovery Score improvement.
- User can approve or reject the schedule patch.
- Demo can be completed within 2-3 minutes.

25. Final Positioning Statement

FlowMate is not built for people whose days are already perfect. FlowMate is built for students whose day has already crashed, but who still want to save what matters.

The unique value is simple:

FlowMate does not just plan your day. It debugs your day.